Course Code	21CSC101T	Course Name	OBJECT ORIENTED DESIGN AND PROGRAMMING	Course Category	C	PROFESSIONAL CORE	L	T	P	C
							2	1	0	3

Pre-requisite Courses	*Nil*	Co- requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*School of Computing*	Data Book / Codes / Standards	*Nil*		

Course Learning Rationale (CLR):	***The purpose of learning this course is to:***
CLR-1:	*programs using object-oriented approach and design methodologies for real-time application development*
CLR-2:	*method overloading and operator overloading for real-time application development programs*
CLR-3:	*inline, friend and virtual functions and create application development programs*
CLR-4:	*exceptional handling and collections for real-time object-oriented programming applications*
CLR-5:	*model the System using Unified Modelling approach using different diagrams*

Course Outcomes (CO):	***At the end of this course, learners will be able to:***	1 Engineering Knowledge	2 Problem Analysis	3 Design/development of solutions	4 Conduct investigations of complex problems	5 Modern Tool Usage	6 The engineer and society	7 Environment & Sustainability	8 Ethics	9 Individual & Team Work	10 Communication	11 Project Mgt. & Finance	12 Life Long Learning	PSO-1	PSO-2	PSO-3
CO-1:	*create programs using object-oriented approach and design methodologies*	-	2	2	-	2	-	-	-	-	-	-	3	-	-	-
CO-2:	*construct programs using method overloading and operator overloading*	-	2	2	-	2	-	-	-	-	-	-	3	-	-	-
CO-3:	*create programs using inline, friend and virtual functions, construct programs using standard templates*	-	2	2	-	2	-	-	-	-	-	-	3	-	-	-
CO-4:	*construct programs using exceptional handling and collections*	-	2	2	-	2	-	-	-	-	-	-	3	-	-	-
CO-5:	*create Models of the system using UML Diagrams*	-	2	2	-	2	-	-	-	-	-	-	3	-	-	-

(Program Outcomes (PO): columns 1–12; Program Specific Outcomes: PSO-1, PSO-2, PSO-3)

Unit-1 - Introduction to OOPS — ***9 Hour***

Object-Oriented Programming - Features of C++ - I/O Operations, Data Types, Variables-Static, Constants-Pointers-Type Conversions – Conditional and looping statements – Arrays - C++ 11 features - Class and Objects, Abstraction and Encapsulation, Access Specifiers, Methods- UML Diagrams Introduction – Use Case Diagram - Class Diagram.

Unit-2 - Methods and Polymorphism — ***9 Hour***

Constructors- Types of constructors - Static constructor and Copy constructor -Destructor - Polymorphism: Constructor overloading - Method Overloading Operator Overloading - UML Interaction Diagrams -Sequence Diagram - Collaboration Diagram - Example Diagram

Unit-3 - Inheritance — ***9 Hour***

Inheritance – Types -Single and Multiple Inheritance - Multilevel Inheritance - Hierarchical Inheritance - Hybrid Inheritance - Advanced Functions - Inline, Friend- Virtual - Pure Virtual function - Abstract class - UML State Chart Diagram - UML Activity Diagram

Unit-4 - Generic Programming — ***9 Hour***

Generic - Templates - Function templates - Class Templates - Exceptional Handling: try and catch - Multilevel exceptional - throw and throws - finally - User defined exceptional - Dynamic Modeling: Package Diagram - UML Component Diagram - UML Deployment Diagram

Unit-5 - Standard Template Library — ***9 Hour***

STL: Containers: Sequence and Associative Container - Sequence Container: Vector, List, Deque, Array, Stack - Associative Containers: Map, Multimap - Iterator and Specialized iterator - Functions of iterator - Algorithms: find(), count(), sort() - Algorithms: search(), merge(), for_each(), transform()

Learning Resources	1. Grady Booch, Robert A. Maksimchuk, Michael W. Engle, Object-Oriented Analysis and Design with Applications, 3rd ed., Addison-Wesley, May 2007 2. Reema Thareja, Object Oriented Programming with C++, 1st ed., Oxford University Press, 2015 3. Sourav Sahay, Object Oriented Programming with C++, 2nd ed., Oxford University Press, 2017	4. Robert Lafore, Object-Oriented Programming in C++, 4th ed., SAMS Publishing, 2008 5. Ali Bahrami, Object Oriented Systems Development", McGraw Hill, 2004 6. Craig Larmen, Applying UML and Patterns, 3rd ed., Prentice Hall, 2004

Learning Assessment

	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (50%)		Life-Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	20%	-	20%	-
Level 2	Understand	20%	-	20%	-	20%	-
Level 3	Apply	30%	-	30%	-	30%	-
Level 4	Analyze	30%	-	30%	-	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers

Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
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